

① $\log_2(5-x) - \log_2(x-2) = 3 - \log_2(1+x)$
 $\log_2(5-x) - \log_2(x-2) + \log_2(1+x) = 3$
 $\log_2\left[\frac{(5-x)(1+x)}{x-2}\right] = 3$ (K1) rule
 $\frac{(5-x)(1+x)}{x-2} = 2^3$ (K1)
 $(5-x)(1+x) = 8(x-2)$
 $5 + 5x - x - x^2 = 8x - 16$
 $x^2 + 4x - 21 = 0$ (K1) quadratic
 $(x+7)(x-3) = 0$ (K1) factorize
 $x+7=0$ $x-3=0$
 $x=-7$ $x=3$ (J1) both
 $\therefore x=3$ (J1)

[6m]

② $\frac{1}{2x-1} < \frac{1}{x+2}$
 $\frac{1}{2x-1} - \frac{1}{x+2} < 0$ (K1)
 $\frac{x+2 - (2x-1)}{(2x-1)(x+2)} < 0$ (K1)
 $\frac{-x+3}{(2x-1)(x+2)} < 0$ (J1)

critical number:

$-x+3=0$ $2x-1=0$ $x+2=0$
 $x=3$ $x=\frac{1}{2}$ $x=-2$

interval	$-\infty, -2$	$-2, \frac{1}{2}$	$\frac{1}{2}, 3$	$3, \infty$
value x	-3	0	1	4
$-x+3$	+	+	+	-
$2x-1$	-	-	+	+
$x+2$	-	+	+	+
$\frac{-x+3}{(2x-1)(x+2)}$	+	-	+	-

(K1) table of sign
(J1) correct sign

$\therefore \{x: -2 < x < \frac{1}{2} \text{ or } x > 3\}$ (J1)

[6m]

③ $z = 1 - \sqrt{3}i$, $\bar{z} = 1 + \sqrt{3}i$

$z^2 = (1 - \sqrt{3}i)^2$
 $z^2 = 1 - 2\sqrt{3}i + 3i^2$ (K1)
 $z^2 = 1 - 2\sqrt{3}i + 3(-1)$ (K1) $i^2 = -1$
 $z^2 = -2 - 2\sqrt{3}i$ (J1)
 $\bar{z}^2 = -2 + 2\sqrt{3}i$ (J1)

$\bar{z}^2 = k \frac{1}{z}$

$-2 + 2\sqrt{3}i = k \left(\frac{1}{1 + \sqrt{3}i} \right)$ (K1)

$k = (-2 + 2\sqrt{3}i)(1 + \sqrt{3}i)$
 $k = -2 - 2\sqrt{3}i + 2\sqrt{3}i + 2(3)i^2$ (K1) simplify
 $k = -2 + 2(3)(-1)$
 $k = -8$ (J1)

[7m]

④ a) $|x^2 - x - 3| = 3$

$$\begin{aligned} x^2 - x - 3 &= 3 & -(x^2 - x - 3) &= 3 & \textcircled{KI} & \text{both} \\ x^2 - x - 6 &= 0 & x^2 - x - 3 &= -3 & \textcircled{KI} & \text{quadratic} \\ (x-3)(x+2) &= 0 & x^2 - x &= 0 & & \\ x=3, x=-2 & & x(x-1) &= 0 & \textcircled{KI} & \text{factorize} \\ & & x=0, x=1 & & \textcircled{JI} & 2 \text{ correct} \\ \therefore x &= -2, 0, 1, 3 & \textcircled{JI} & \text{all correct} & & \\ & & & & & [5m] \end{aligned}$$

b) $\frac{2x^2 + 9x - 4}{x+2} < 4$

$$\frac{2x^2 + 9x - 4}{x+2} - 4 < 0 \quad \textcircled{KI}$$

$$\frac{2x^2 + 9x - 4 - 4(x+2)}{x+2} < 0 \quad \textcircled{KI} \text{ equate denom}$$

$$\frac{2x^2 + 5x - 12}{x+2} < 0 \quad \textcircled{KI} \text{ quadratic numerator}$$

$$\frac{(2x-3)(x+4)}{x+2} < 0 \quad \textcircled{JI}$$

critical number: $x = \frac{3}{2}, x = -4, x = -2$

interval	$-\infty, -4$	$-4, -2$	$-2, \frac{3}{2}$	$\frac{3}{2}, \infty$
value x	-5	-3	0	2
$2x-3$	-	-	-	+
$x+4$	-	+	+	+
$x+2$	-	-	+	+
$\frac{(2x-3)(x+4)}{x+2}$	$\ominus$	+	$\ominus$	+

④ KI

④ JI

solution set is $\{x: x < -4 \cup -2 < x < \frac{3}{2}\} \quad \textcircled{JI}$

[7m]

⑤ a) $f(x) = \ln(2x+3)$

$$\begin{aligned} f(x_1) &= f(x_2) \\ \ln(2x_1+3) &= \ln(2x_2+3) & (K1) \\ 2x_1+3 &= 2x_2+3 \\ 2x_1 &= 2x_2 \\ x_1 &= x_2 & (K1) \end{aligned}$$

Since $x_1 = x_2$, therefore $f(x)$ is a one-to-one function. (J1)

[3m]

b) $f(g(x)) = \ln\left(2\left(\frac{e^x-3}{2}\right)+3\right)$ (K1)

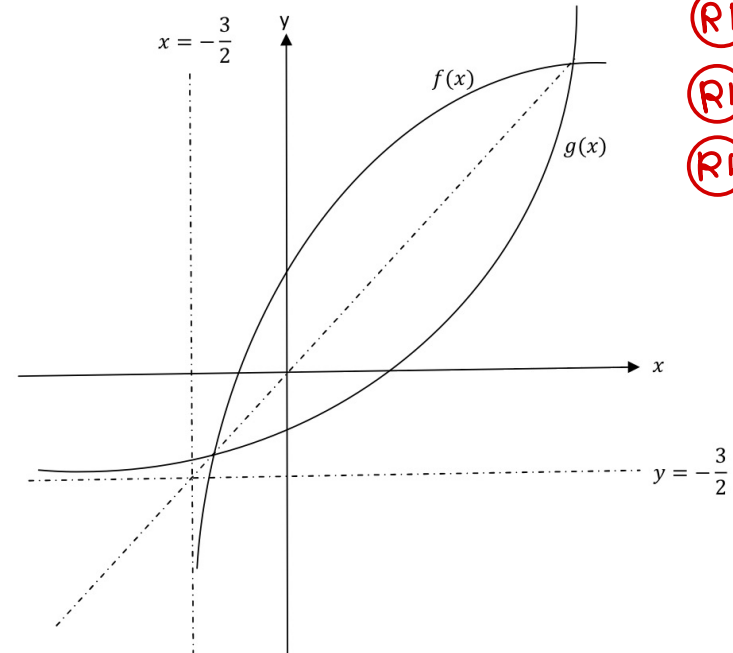
$$\begin{aligned} &= \ln(e^x - 3 + 3) \\ &= \ln e^x \\ &= x & (J1) \end{aligned}$$

$$\begin{aligned} g(f(x)) &= \frac{e^{\ln(2x+3)} + 3}{2} & (K1) \\ &= \frac{2x+3+3}{2} \end{aligned}$$

$$g(f(x)) = x \quad (J1)$$

Since $f(g(x)) = g(f(x)) = x$, therefore $f(x)$ and $g(x)$ are inverses to each other. (J1)

[5m]



(R1) $f(x)$

(R1) $g(x)$

(R1) all correct with label

$$D_f = \left(-\frac{3}{2}, \infty\right) \quad (B1)$$

$$R_f = (-\infty, \infty) \quad (B1)$$

[5m]

$$\textcircled{6} \quad f(x) = \begin{cases} 1+e^x, & x < 1 \\ 1, & x = 1 \\ 2-x, & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 1+e^x = 1+e^1 = 1+e \quad (\text{KI}) \quad (\text{JI})$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 2-x = 2-1 = 1 \quad (\text{KI}) \quad (\text{JI})$$

Since $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$, therefore

$\lim_{x \rightarrow 1} f(x)$ does not exist. (JI)

[5m]

$$\textcircled{7} \quad \text{a) } \lim_{x \rightarrow \infty} \frac{2x^2 + x - 4}{1 - x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2} + \frac{x}{x^2} - \frac{4}{x^2}}{\frac{1}{x^2} - \frac{x^2}{x^2}} \quad (\text{KI})$$

$$= \lim_{x \rightarrow \infty} \frac{2 + \frac{1}{x} - \frac{4}{x^2}}{\frac{1}{x^2} - 1}$$

$$= \frac{2 + 0 - 0}{0 - 1} \quad (\text{KI})$$

$$= -2 \quad (\text{JI})$$

[3m]

$$\text{b) } \lim_{x \rightarrow 2} \frac{3 - \sqrt{x+7}}{x^2 - 4}$$

$$= \lim_{x \rightarrow 2} \frac{3 - \sqrt{x+7}}{x^2 - 4} \times \frac{3 + \sqrt{x+7}}{3 + \sqrt{x+7}} \quad (\text{KI}) \quad \text{conjugate}$$

$$= \lim_{x \rightarrow 2} \frac{9 - (x+7)}{(x^2 - 4)(3 + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{2 - x}{(x-2)(x+2)(3 + \sqrt{x+7})} \quad (\text{KI}) \quad \text{simplify}$$

$$= \lim_{x \rightarrow 2} \frac{-(x-2)}{(x-2)(x+2)(3 + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{-1}{(x+2)(3 + \sqrt{x+7})}$$

$$= \frac{-1}{(2+2)(3 + \sqrt{2+7})} \quad (\text{KI})$$

$$= -\frac{1}{24} \quad (\text{JI})$$

[4m]

$$\textcircled{8} \text{ a) } f(x) = \begin{cases} \frac{|x^2 - x - 2|}{x^2 - 2x} & , \quad x \neq 0, 2 \\ 0 & , \quad x = 2 \end{cases}$$

$$|x^2 - x - 2| = \begin{cases} x^2 - x - 2 & , \quad x < -1 \text{ or } x > 2 \\ -(x^2 - x - 2) & , \quad -1 < x < 2 \end{cases}$$

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} \frac{-(x^2 - x - 2)}{x^2 - 2x} \\ &= \lim_{x \rightarrow 2^-} \frac{-(x+1)(x-2)}{x(x-2)} \quad \textcircled{KI} \\ &= \lim_{x \rightarrow 2^-} \frac{-(x+1)}{x} \\ &= -\frac{(2+1)}{2} \\ &= -\frac{3}{2} \quad \textcircled{JI} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^+} \frac{x^2 - x - 2}{x^2 - 2x} \\ &= \lim_{x \rightarrow 2^+} \frac{(x+1)(x-2)}{x(x-2)} \quad \textcircled{KI} \\ &= \lim_{x \rightarrow 2^+} \frac{(x+1)}{x} \\ &= \frac{2+1}{2} \\ &= \frac{3}{2} \quad \textcircled{JI} \end{aligned}$$

$\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$. Thus $\lim_{x \rightarrow 2} f(x)$ does not exist. $\textcircled{JI}$

So, $f(x)$ is not continuous at $x=2$. $\textcircled{JI}$

[6m]

$$\text{b) } f(x) = \begin{cases} \alpha x + 6 & , \quad x < 4 \\ x^2 + 2 & , \quad 4 \leq x < 6 \\ 2 - \beta x & , \quad x \geq 6 \end{cases}$$

$f(x)$ is continuous,

$$\begin{aligned} f(4) &= \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x) \\ 4^2 + 2 &= \lim_{x \rightarrow 4^-} \alpha x + 6 = \lim_{x \rightarrow 4^+} x^2 + 2 \quad \textcircled{BI} \end{aligned}$$

$$\begin{aligned} 4^2 + 2 &= \lim_{x \rightarrow 4^-} \alpha x + 6 \\ 18 &= \alpha(4) + 6 \quad \textcircled{KI} \\ 4\alpha &= 12 \\ \alpha &= 3 \end{aligned}$$

$$\begin{aligned} f(6) &= \lim_{x \rightarrow 6^-} x^2 + 2 = \lim_{x \rightarrow 6^+} 2 - \beta x \\ 2 - \beta(6) &= \lim_{x \rightarrow 6^-} x^2 + 2 = \lim_{x \rightarrow 6^+} 2 - \beta x \quad \textcircled{BI} \end{aligned}$$

$$\begin{aligned} 2 - 6\beta &= 6^2 + 2 \quad \textcircled{KI} \\ 2 - 6\beta &= 38 \\ -6\beta &= 36 \\ \beta &= -6 \end{aligned}$$

$\therefore \alpha = 3, \beta = -6$ $\textcircled{JI}$ both

[5m]

9 a) $y = \frac{1}{\sqrt{2x+1}}$

i. let $f(x) = \frac{1}{\sqrt{2x+1}}$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{1}{\sqrt{2(x+h)+1}} - \frac{1}{\sqrt{2x+1}} \right] \quad (K1) \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{\sqrt{2x+1} - \sqrt{2(x+h)+1}}{\sqrt{2(x+h)+1} \sqrt{2x+1}} \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{\sqrt{2x+1} - \sqrt{2(x+h)+1}}{\sqrt{2(x+h)+1} \sqrt{2x+1}} \times \frac{\sqrt{2x+1} + \sqrt{2(x+h)+1}}{\sqrt{2x+1} + \sqrt{2(x+h)+1}} \right] \quad (K1) \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{2x+1 - (2(x+h)+1)}{\sqrt{2(x+h)+1} \sqrt{2x+1} (\sqrt{2x+1} + \sqrt{2(x+h)+1})} \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{-2h}{\sqrt{2(x+h)+1} \sqrt{2x+1} (\sqrt{2x+1} + \sqrt{2(x+h)+1})} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{-2}{\sqrt{2(x+h)+1} \sqrt{2x+1} (\sqrt{2x+1} + \sqrt{2(x+h)+1})} \right] \\
 &= \frac{-2}{\sqrt{2(x+0)+1} \sqrt{2x+1} (\sqrt{2x+1} + \sqrt{2(x+0)+1})} \quad (K1) \\
 &= \frac{-2}{\sqrt{2x+1} \sqrt{2x+1} (2\sqrt{2x+1})} \\
 &= \frac{-1}{(2x+1)^{3/2}} \quad (J1)
 \end{aligned}$$

[4m]

ii. $\frac{dy}{dx} = -(2x+1)^{-3/2}$

$$\frac{d^2y}{dx^2} = \frac{3}{2} (2x+1)^{-5/2} (2) \quad (K1)$$

$$\frac{d^2y}{dx^2} = \frac{3}{(2x+1)^{5/2}} \quad (J1)$$

[2m]

b) i. $y = e^{2x} \tan x$

$$\begin{aligned}
 u &= e^{2x} & v &= \tan x \\
 u' &= 2e^{2x} & v' &= \sec^2 x
 \end{aligned}$$

$$\frac{dy}{dx} = \tan x (2e^{2x}) + e^{2x} (\sec^2 x) \quad (K1)$$

$$\frac{dy}{dx} = e^{2x} (2 \tan x + \sec^2 x) \quad (J1)$$

[2m]

$$\begin{aligned} \text{ii. } y &= x^{\sec x} \\ \ln y &= \ln x^{\sec x} \\ \ln y &= \sec x \ln x \end{aligned}$$

$$\begin{aligned} u &= \sec x & v &= \ln x \\ u' &= \sec x \tan x & v' &= \frac{1}{x} \end{aligned} \quad \text{(K1) } u' \& v'$$

$$\frac{1}{y} \left(\frac{dy}{dx} \right) = \ln x (\sec x \tan x) + \sec x \left(\frac{1}{x} \right) \quad \text{(K1) left + (K1) right}$$

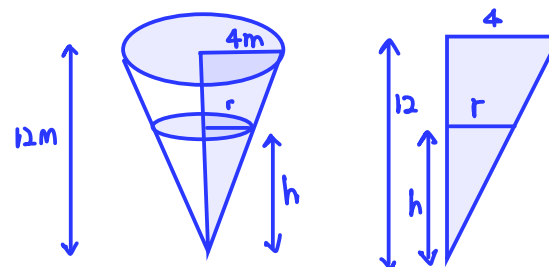
$$\frac{dy}{dx} = y \left[\ln x (\sec x \tan x) + \frac{\sec x}{x} \right]$$

$$\frac{dy}{dx} = x^{\sec x} \left[\ln x (\sec x \tan x) + \frac{\sec x}{x} \right]$$

$$\frac{dy}{dx} = x^{\sec x} \sec x \left[\frac{x \ln x \tan x + 1}{x} \right] \quad \text{(J1)}$$

[4m]

(10) a)



$$\frac{dv}{dt} = 50 \text{ m}^3 \quad \frac{dh}{dt} \quad \text{when } h=6$$

$$\frac{dh}{dt} = \frac{dh}{dv} \times \frac{dv}{dt}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \left(\frac{h}{3} \right)^2 h$$

$$V = \frac{1}{27} \pi h^3 \quad \text{(K1)}$$

$$\frac{dv}{dh} = \frac{1}{9} \pi h^2 \quad \text{(K1)}$$

$$\frac{dh}{dv} = \frac{9}{\pi h^2}$$

$$\frac{dh}{dt} = \frac{9}{\pi h^2} \times 50 \quad \text{(K1)}$$

$$\frac{dh}{dt} = \frac{450}{\pi h^2}$$

$$\begin{aligned} r &= 4 & r &= \frac{4}{12} h \\ h &= 12 & r &= \frac{h}{3} \end{aligned} \quad \text{(K1)}$$

$$\begin{aligned} \text{when } h &= 6 \\ \frac{dh}{dt} &= \frac{450}{\pi (6)^2} \quad \text{(K1)} \\ \frac{dh}{dt} &= \frac{25}{2\pi} \text{ m/min} \quad \text{(J1)} \end{aligned}$$

[6m]

b) surface area cylinder,
 $S = 2\pi r^2 + 2\pi rh$

cost of material cylinder,
 $C = 2\pi r^2(2) + 2\pi rh(1)$
 $C = 4\pi r^2 + 2\pi rh$

$$V = \pi r^2 h \quad (V \text{ is constant})$$
$$h = \frac{V}{\pi r^2} \quad \textcircled{BI}$$

$$C = 4\pi r^2 + 2\pi r h$$
$$= 4\pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) \quad \textcircled{KI}$$

$$C = 4\pi r^2 + \frac{2V}{r}$$

$$\frac{dC}{dr} = 8\pi r - \frac{2V}{r^2} \quad \textcircled{KI} \quad \frac{dC}{dr} = 0$$

$$\frac{dC}{dr} = 0$$

$$8\pi r - \frac{2V}{r^2} = 0$$

$$8\pi r^3 = 2V$$

$$r = \left(\frac{V}{4\pi} \right)^{1/3} \quad \textcircled{JI}$$

$$\frac{d^2C}{dr^2} = 8\pi + \frac{4V}{r^3} \quad \textcircled{KI}$$

$$\text{when } r = \left(\frac{V}{4\pi} \right)^{1/3}$$

$$\frac{d^2C}{dr^2} = 8\pi + \frac{4V}{\left[\left(\frac{V}{4\pi} \right)^{1/3} \right]^3}$$

$$\frac{d^2C}{dr^2} = 8\pi + 16\pi$$

$$\frac{d^2C}{dr^2} = 24\pi > 0 \quad (\text{min}) \quad \textcircled{KI}$$

$$\text{when } r = \left(\frac{V}{4\pi} \right)^{1/3}, \quad h = \frac{V}{\pi \left(\frac{V}{4\pi} \right)^{2/3}} = \left(\frac{16V}{\pi} \right)^{1/3}$$

$\therefore$ the cost is minimum $r = \left(\frac{V}{4\pi} \right)^{1/3}, \quad h = \left(\frac{16V}{\pi} \right)^{1/3} \quad \textcircled{JI} \text{ both}$

$$\textcircled{11} \text{ a) i. } 3y^2 - xy + x^2 = 3$$

$$6y \frac{dy}{dx} - \left[y + x \frac{dy}{dx} \right] + 2x = 0 \quad \textcircled{K1}$$

$$6y \frac{dy}{dx} - y - x \frac{dy}{dx} + 2x = 0$$

$$6y \frac{dy}{dx} - x \frac{dy}{dx} = y - 2x \quad \textcircled{K1}$$

$$(6y - x) \frac{dy}{dx} = y - 2x$$

$$\frac{dy}{dx} = \frac{y - 2x}{6y - x} \quad \textcircled{J1}$$

when $x=1$,

$$3y^2 - 1(y) + 1^2 = 3$$

$$3y^2 - y - 2 = 0$$

$$(3y+2)(y-1) = 0$$

$$y = -\frac{2}{3}, y = 1$$

} $\textcircled{K1}$ find y

when $x=1, y = -\frac{2}{3}$,

$$\frac{dy}{dx} = \frac{-\frac{2}{3} - 2(1)}{6(-\frac{2}{3}) - 1} = \frac{8}{15}$$

when $x=1, y=1$

$$\frac{dy}{dx} = \frac{1 - 2(1)}{6(1) - 1} = -\frac{1}{5}$$

} $\textcircled{K1}$ sub both
 $\textcircled{J1}$ answer both

[6m]

$$\text{a) ii. } (6y - x) \frac{dy}{dx} = y - 2x$$

$$u = 6y - x$$

$$u' = 6 \frac{dy}{dx} - 1$$

$$v = \frac{dy}{dx}$$

$$v' = \frac{d^2y}{dx^2}$$

$$\frac{dy}{dx} \left(6 \frac{dy}{dx} - 1 \right) + (6y - x) \frac{d^2y}{dx^2} = \frac{dy}{dx} - 2 \quad \textcircled{K1} \text{ diff}$$

$$6 \left(\frac{dy}{dx} \right)^2 - \frac{dy}{dx} + (6y - x) \frac{d^2y}{dx^2} - \frac{dy}{dx} + 2 = 0$$

$$(6y - x) \frac{d^2y}{dx^2} + 6 \left(\frac{dy}{dx} \right)^2 - 2 \frac{dy}{dx} + 2 = 0 \quad \textcircled{J1} \text{ shown}$$

[2m]

b) i. $x = 3t - \frac{2}{t}$

$$\frac{dx}{dt} = 3 + \frac{2}{t^2}$$

$$\frac{dx}{dt} = \frac{3t^2 + 2}{t^2} \quad \checkmark$$

$y = 3t + \frac{2}{t}$

$$\frac{dy}{dt} = 3 - \frac{2}{t^2}$$

$$\frac{dy}{dt} = \frac{3t^2 - 2}{t^2} \quad \checkmark \quad \textcircled{\text{KI}} \text{ both}$$

$$\frac{dy}{dx} = \frac{3t^2 - 2}{t^2} \times \frac{t^2}{3t^2 + 2} \quad \textcircled{\text{KI}}$$

$$\frac{dy}{dx} = \frac{3t^2 - 2}{3t^2 + 2}$$

$$\begin{array}{r} 1 \\ 3t^2 + 2 \overline{) 3t^2 - 2} \\ \hline \hookrightarrow \frac{3t^2 + 2}{-4} \end{array}$$

$$\frac{dy}{dx} = 1 - \frac{4}{3t^2 + 2}$$

shown $\textcircled{\text{JI}}$

[3m]

ii. $\frac{d^2y}{dx^2} = 4(3t^2 + 2)^{-2}(6t) \times \frac{t^2}{3t^2 + 2} \quad \textcircled{\text{KI}}$

$$\frac{d^2y}{dx^2} = \frac{24t}{(3t^2 + 2)^2} \times \frac{t^2}{3t^2 + 2}$$

$$\frac{d^2y}{dx^2} = \frac{24t^3}{(3t^2 + 2)^3} \quad \textcircled{\text{JI}}$$

when $t=1$,

$$\frac{d^2y}{dx^2} = \frac{24(1)^3}{(3(1)^2 + 2)^3} \quad \textcircled{\text{KI}}$$

$$= \frac{24}{125} \quad \textcircled{\text{JI}}$$

[4m]